

XRD peak fitting for Nb films with different stress

Compiled data

Peak position (2-theta) in degrees, all to 5 d.p.

- April has lower signal in data due to it being a spot check
- June JT 4 data was corrupt / overwritten, we decided not to retake back then as data was deemed sufficient

Film	20250421	20250611
JT1	38.03579 +/- 0.00128	38.03590 +/- 0.00068
JT2	38.22767 +/- 0.00296	38.22585 +/- 0.00073
JT3	38.13615 +/- 0.00226	38.16354 +/- 0.00097
JT4	38.41943 +/- 0.00475	NA
JT5	38.24877 +/- 0.00422	38.26104 +/- 0.00086
JT6	38.26100 +/- 0.00189	38.27461 +/- 0.00058
JT7	38.26570 +/- 0.00104	38.26719 +/- 0.00039
JT8	38.16273 +/- 0.00223	38.16516 +/- 0.00094
JT9	38.10819 +/- 0.00126	38.11518 +/- 0.00053
JT10	38.20222 +/- 0.00076	38.20624 +/- 0.00036

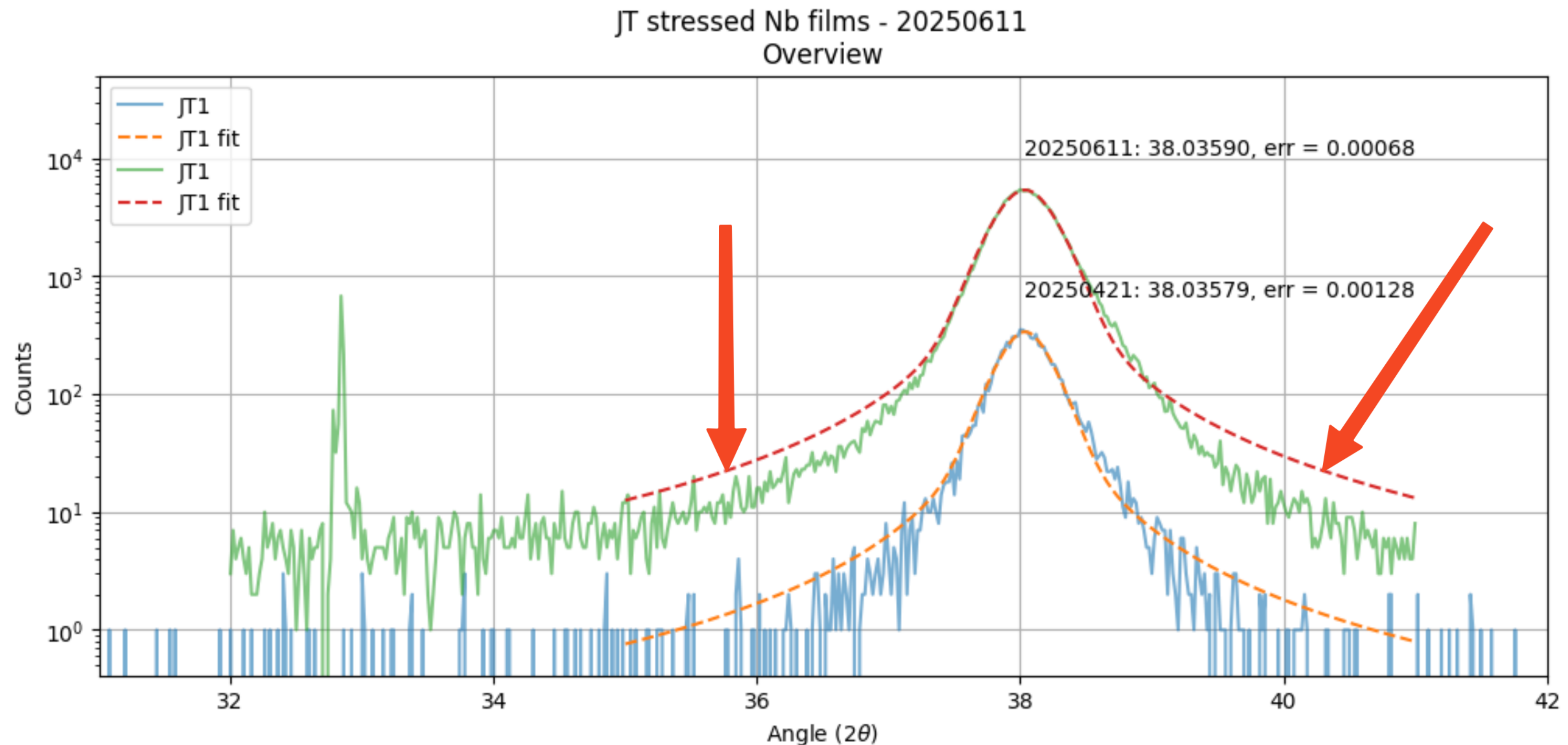
Methodology: Pseudo Voigt Function

- In the ideal case the fit follows a convolution of Gaussian and Lorentzian due to different scattering mechanisms
- As a simplified approximation for easier fitting, I used a pseudo Voigt function, a weighted average of a Gaussian and Lorentzian:

```
# Pseudo-Voigt function  
def pseudo_voigt(x, A, x0, sigma, gamma, eta):  
    gaussian = A * np.exp(-(x - x0)**2 / (2 * sigma**2))  
    lorentzian = A * gamma**2 / ((x - x0)**2 + gamma**2)  
    return eta * lorentzian + (1 - eta) * gaussian
```


Outstanding issues (example)

Fit still not the best away from peak, but I do not believe this affects peak values



Compiled data with FWHM

Peak position (2-theta) in degrees, all to 5 d.p.

Film	20250421	FWHM	20250611	FWHM
JT1	38.03579 +/- 0.00128	0.44764	38.03590 +/- 0.00068	0.51665
JT2	38.22767 +/- 0.00296	0.51065	38.22585 +/- 0.00073	0.57486
JT3	38.13615 +/- 0.00226	0.49985	38.16354 +/- 0.00097	0.58086
JT4	38.41943 +/- 0.00475	0.50105	NA	NA
JT5	38.24877 +/- 0.00422	0.59886	38.26104 +/- 0.00086	0.60306
JT6	38.26100 +/- 0.00189	0.48425	38.27461 +/- 0.00058	0.56166
JT7	38.26570 +/- 0.00104	0.40504	38.26719 +/- 0.00039	0.49505
JT8	38.16273 +/- 0.00223	0.52865	38.16516 +/- 0.00094	0.58866
JT9	38.10819 +/- 0.00126	0.42664	38.11518 +/- 0.00053	0.52265
JT10	38.20222 +/- 0.00076	0.38704	38.20624 +/- 0.00036	0.47285

Error estimation using repeatability

- Using the 9 films which has two measurements, we compute difference between the two measurements and find the standard deviation

```
peak_Apr = [38.03579, 38.22767, 38.13615, 38.24877, 38.26100, 38.26570, 38.16273, 38.10819, 38.20222]  
peak_Jun = [38.03590, 38.22585, 38.16354, 38.26104, 38.27461, 38.26719, 38.16516, 38.11518, 38.20624]  
differences = np.array(peak_Apr) - np.array(peak_Jun)  
exp_error = np.std(differences)
```

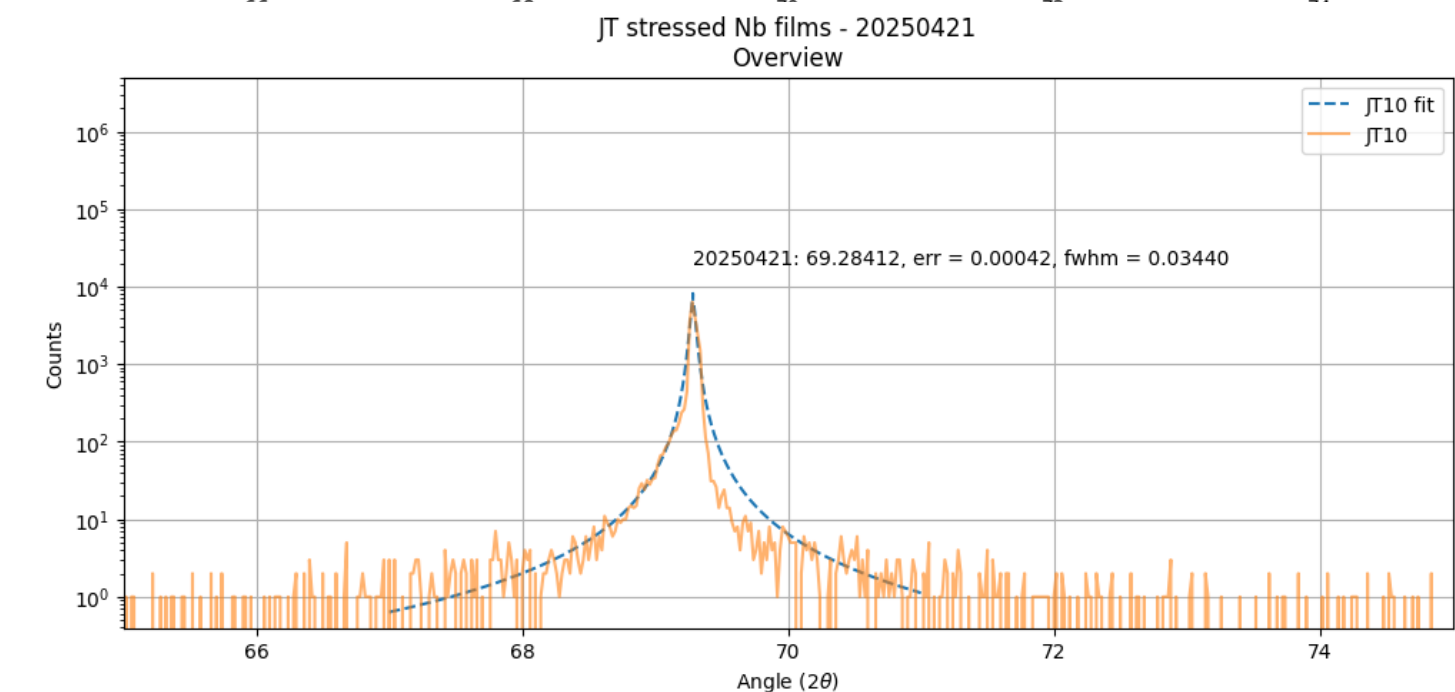
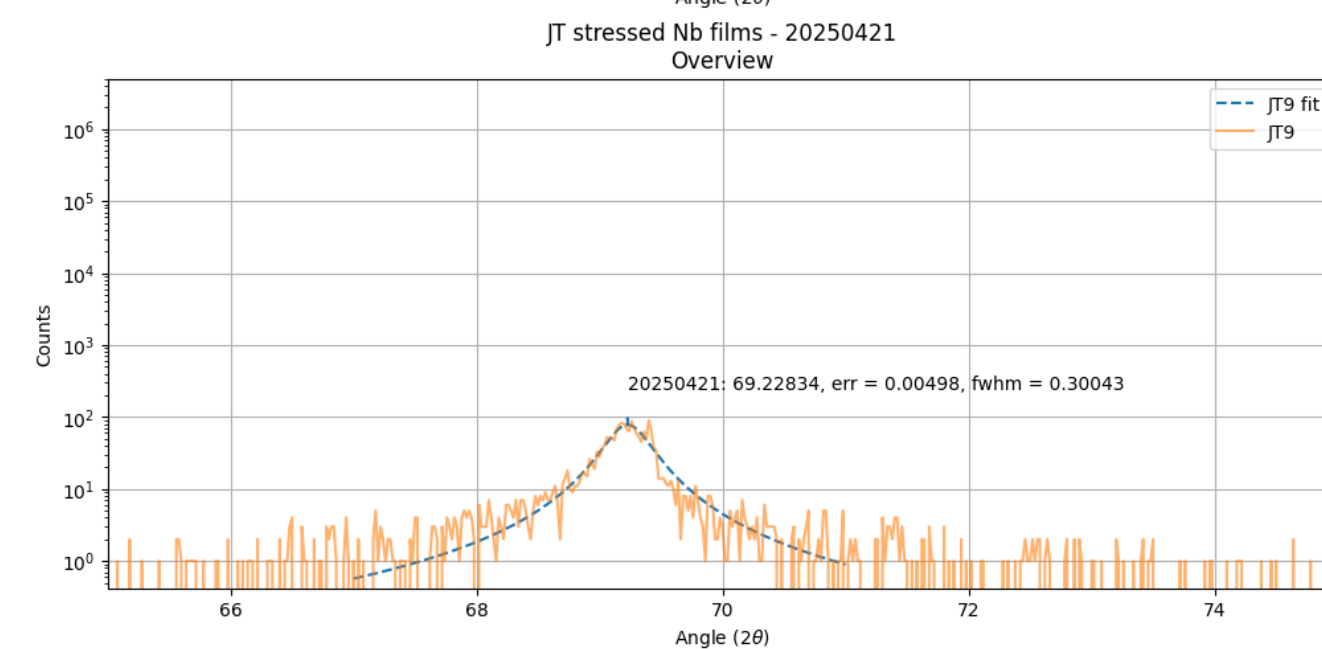
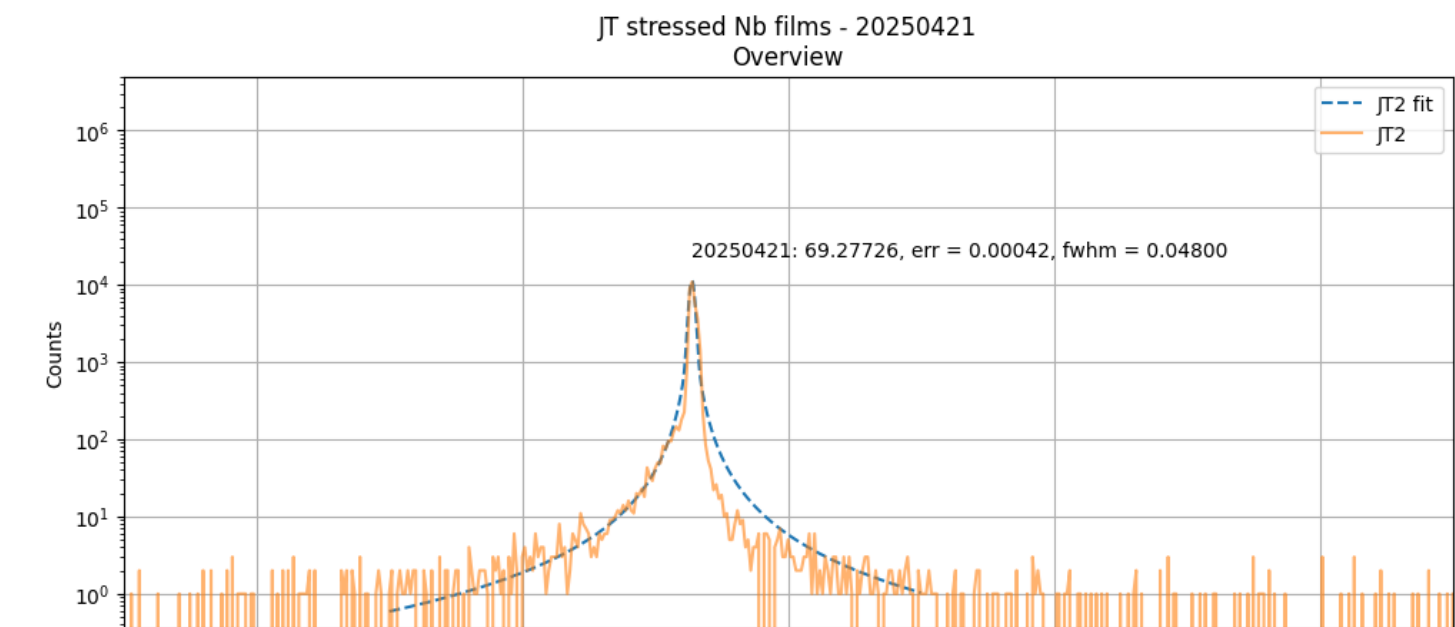
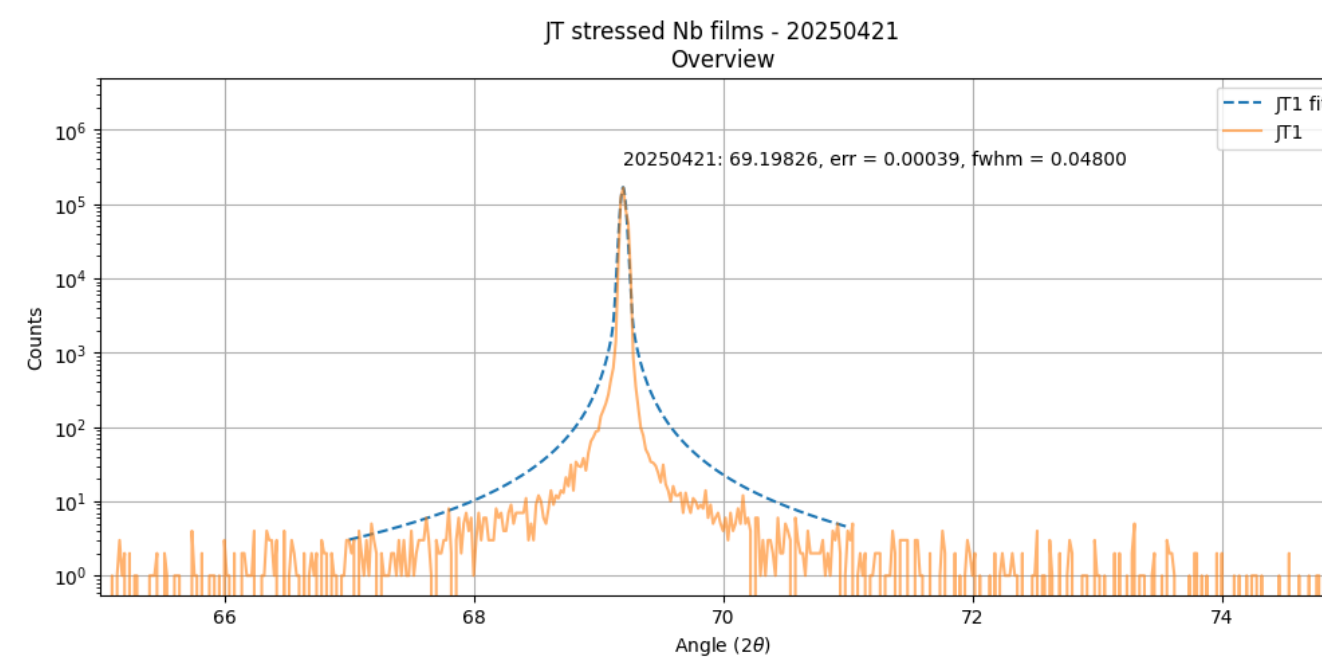
✓ 0.0s

- Gives us error of 0.0086 degrees
- Combined with statistical errors from before we get roughly 0.01 degrees

Error estimation using repeatability - Si

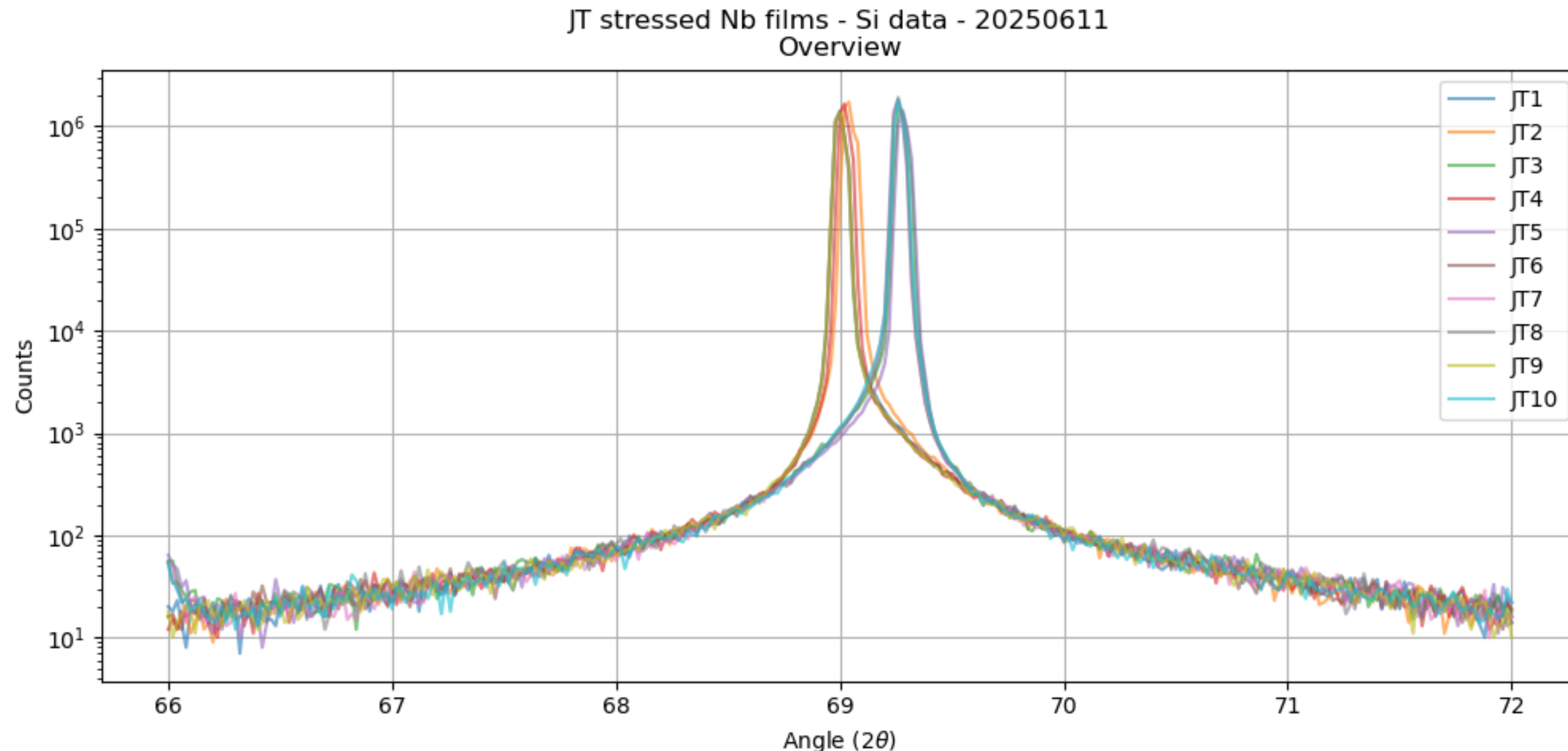
Used Apr scan values for initial attempt

- Scans had Si (002) peaks (literature value theta-2-theta 69.12 deg) so we can do statistics on that too (example fits on right)
- Not perfect fits but peak values should be good
- Statistics
 - Mean = 69.20
 - Std = 0.08
- Largest value of error so far



Issue with using Si peaks: some kind of bimodal distribution?

- I remember bringing this up but this was the June data I took:



This bimodal is not consistent in the measurement in April

- My suspicion is miscut - I may have been more careful with wafer orientation in June, e.g. flat facing 1 side consistently, but I think Si isn't good for checking errors. Maybe we can stick to the Nb-repeatability

